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ORIGINAL RESEARCH

Cyclic codes over $\mathcal{M}_4(\mathbb{F}_2)$

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Abstract
In this article, keeping the huge research prospective of the study in mind, we consider the non-commutative ring $\mathcal{M}_4(\mathbb{F}_2)$, the set of all 4×4 matrices over the field $\mathbb{F}_2$ and confirm that this ring is isomorphic with the ring $\mathbb{F}_{16} + u\mathbb{F}_{16} + u^2\mathbb{F}_{16} + u^3\mathbb{F}_{16}$, where $u^4 = 0$. Besides, we develop the structure of cyclic codes and their generators over the ring. Also, making use of Gray map from $\mathcal{M}_4(\mathbb{F}_2)$ to $\mathbb{F}_{16}^4$, we infer that the image of a cyclic code is a linear code. Finally, our findings are authenticated by suitable non-trivial examples.

Keywords Matrix ring · Modules · Cyclic codes · Quaternary codes

Mathematics Subject Classification 94B05 · 94B15

1 Introduction

Cyclic codes over finite rings have been extensively studied due to their vast significance in the theory of error correcting codes. Besides, codes over finite rings have experienced a remarkable growth after the development of linear codes over the ring of integers modulo 4 obtained in [7]. In that article, the authors gave a new wing to the topic by employing Gray map from $\mathbb{Z}_4$ to $\mathbb{F}_2^2$ which relates some illustrious non-linear codes to some linear codes. Since then, codes have been thoroughly studied over finite commutative rings by several mathematicians and consequently plenty of interesting results on codes over finite commutative rings are there in the literature, among those some of are related to chain [1,4,6,10] and some others are non-chain [13,14]. A very

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few authors work on non-commutative rings to explore a new direction in coding theory. However, structure of cyclic codes and their duals over non-commutative finite rings are new in coding theory [2,3,5,8,9,11].

In 2013, Alahmadi et al. [2] characterized cyclic codes and their duals over the matrix ring, namely $\mathcal{M}_2(\mathbb{F}_2)$. The Bachoc maps from the ring to $\mathbb{F}_4^2$, two copies of Galois field $\mathbb{F}_4$ gives formally self-dual $\mathbb{F}_4$-codes. On the other hand, in their research article, Luo and Udaya [8] derived the structure of cyclic codes over the matrix ring $\mathcal{M}_2(\mathbb{F}_2 + u\mathbb{F}_2)$ and established an isometric map from $\mathcal{M}_2(\mathbb{F}_2 + u\mathbb{F}_2)$ to $\mathbb{F}_4 + u\mathbb{F}_4 + v\mathbb{F}_4 + uv\mathbb{F}_4$ to show some optimal cyclic codes of even length over $\mathbb{F}_4$. Recently, Bhowmick et al. [5] proposed a matrix ring over the ring of characteristic 4, namely $\mathcal{M}_2(\mathbb{Z}_4)$. The authors further achieved that $\mathcal{M}_2(\mathbb{Z}_4) \cong \mathbb{Z}_4[\omega] + U\mathbb{Z}_4[\omega]$, where ω is a root of the polynomial $x^2 + x + 1$, and $U \equiv \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \in \mathcal{M}_2(\mathbb{Z}_4)$. Finally, the structure of cyclic codes and their self-duals have also been investigated over this ring.

In this manuscript, we consider the non-commutative ring $\mathcal{M}_4(\mathbb{F}_2)$, the set of all 4×4 matrices over the field $\mathbb{F}_2$ in [9] and concentrate only on the structure of cyclic codes over such ring. It has been shown that $\mathcal{M}_4(\mathbb{F}_2) \cong \mathbb{F}_{16} + e\mathbb{F}_{16} + e^2\mathbb{F}_{16} + e^3\mathbb{F}_{16}$, where $\mathbb{F}_{16}$ is a field isomorphic to $\mathbb{F}_2[\omega]$, ω is a root of the equation $x^4 + x + 1 = 0$ and e is an indeterminate with $(1+e)^4 = 0$. Moreover, an isometic map from $\mathcal{M}_4(\mathbb{F}_2)$ to $\mathbb{F}_{16}^4$ is defined which preserves minimum weight.

To keep the article self-contained, we put down a brief description of the ring $\mathcal{M}_4(\mathbb{F}_2)$ in Sect. 2. Further, Sect. 3 deals with the structure of cyclic codes over the ring described above and also offers a Gray map to demonstrate a correspondence between $\mathcal{M}_4(\mathbb{F}_2)$ and $\mathbb{F}_{16}^4$.

2 Preliminaries

Throughout this article, we denote the ring alphabet $\mathcal{M}_4(\mathbb{F}_2)$ by $\mathbb{A}$. To begin with, we consider the ring $\mathbb{F}_{16} + e\mathbb{F}_{16} + e^2\mathbb{F}_{16} + e^3\mathbb{F}_{16}$, where e is an indeterminate and $\mathbb{F}_{16}$ is isomorphic to $\mathbb{F}_2[\omega]$, where w satisfies the biquadratic irreducible polynomial $x^4 + x + 1$ over $\mathbb{F}_2$. As in [9], if we consider

$$e = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \text{ and } \omega = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix},$$

the ring $\mathbb{A}$ is isomorphic to the non-commutative ring $\mathbb{F}_{16} + e\mathbb{F}_{16} + e^2\mathbb{F}_{16} + e^3\mathbb{F}_{16}$, where the indeterminate e satisfies the relation $(1 + e)^4 = 0$. Moreover, the ring $\mathbb{F}_{16} + e\mathbb{F}_{16} + e^2\mathbb{F}_{16} + e^3\mathbb{F}_{16}$ can be alternatively represented as $\mathbb{F}_{16} + u\mathbb{F}_{16} + u^2\mathbb{F}_{16} + u^3\mathbb{F}_{16}$ with $u^4 = (1 + e)^4 = 0$, i.e., u is a nilpotent element of nilpotency 4. To be a ring, we define the addition on the ring as usual component-wise addition and the multiplication operation on $\mathbb{F}_{16} + u\mathbb{F}_{16} + u^2\mathbb{F}_{16} + u^3\mathbb{F}_{16}$ is given in Table 1.

Thus we can conclude that $\mathbb{A} \cong \mathbb{F}_{16} + u\mathbb{F}_{16} + u^2\mathbb{F}_{16} + u^3\mathbb{F}_{16}$.

Table 1 Multiplication table

$\cdot$	1	u	u^2	u^3
1	1	u	u^2	u^3
u	u	u^2	u^3	0
u^2	u^2	u^3	0	0
u^3	u^3	0	0	0

A linear code $\mathcal{C}$ of length n over the ring $\mathbb{A}$ is a right $\mathbb{A}$-submodule of $\mathbb{A}^n$. A code $\mathcal{C}$ is called cyclic if for every codeword $x = (x_0, x_1, \ldots, x_{n-1}) \in \mathcal{C}$, its cyclic shift $(x_{n-1}, x_0, x_1, \ldots, x_{n-2})$ is also in $\mathcal{C}$.

Let $\mathcal{R}_n = \frac{\mathbb{A}[x]}{\langle x^n-1\rangle}$. Then any right $\mathbb{A}[x]$-submodule of $\mathcal{R}_n$ represents a cyclic code of length n.

Let us consider a mapping,

$$\mu : \mathbb{A}[x] \to \mathbb{F}_{16}[x] \quad \text{such that}$$
$$\sum_{i=0}^{n-1} a_i x^i \mapsto \sum_{i=0}^{n-1} \bar{\mu}(a_i) x^i,$$

where $\bar{\mu}(a_i)$ is denoted by reduction to modulo u of a_i.

A polynomial is called basic irreducible over $\mathbb{A}$ if it is irreducible over $\mathbb{F}_{16}$. Consider $x^n - 1 = \prod_{r=1}^{m} f_r$, where f_r's are irreducible polynomials over $\mathbb{F}_{16}$ and n is always odd. Then the above polynomial can be uniquely factorized into pairwise coprime polynomials over $\mathbb{F}_{16}$.

3 Main results

Theorem 1 *Let n be an odd positive integer and $x^n - 1 = \prod_{r=1}^{m} f_r$, where f_r's are basic irreducible polynomials in $\mathbb{A}[x]$. Then*

$$\mathcal{R}_n = \frac{\mathbb{A}[x]}{\langle x^n - 1\rangle} = \bigoplus_{r=1}^{m} \frac{\mathbb{A}[x]}{\langle f_r\rangle}.$$

Proof Proof of the theorem is an immediate consequence of [9] and [12]. □

This theorem is entitled as non-commutative analogue of Chinese Remainder Theorem. Since, $\langle f_r\rangle$ is not a two sided ideal, we call $\frac{\mathbb{A}[x]}{\langle f_r\rangle}$ as a right $\mathbb{A}$-module.

Theorem 2 *Let f be a basic irreducible polynomial in $\mathbb{A}[x]$. Then the right $\mathbb{A}$-submodules of $\mathcal{R}_f = \frac{\mathbb{A}[x]}{\langle f\rangle}$ are $\langle 0\rangle$, $\langle 1\rangle$, $\langle u\rangle$, $\langle u^2\rangle$, and $\langle u^3\rangle$.*

Proof Let I be a non-zero right $\mathbb{A}$-submodule of $\mathcal{R}_f$ and $g(x)$ be a non-zero element in I. Consider $g(x) = g_1(x) + ug_2(x) + u^2 g_3(x) + u^3 g_4(x)$, where $g_i(x) \in \mathbb{F}_{16}[x]$,

for $i = 1, 2, 3, 4$. Since, according to our assumption $f(x)$ is irreducible over $\mathbb{F}_{16}$, we have $\gcd(f(x), g_1(x)) = 1$ or $f(x)$. If $\gcd(f(x), g_1(x)) = 1$, then there exists $h_1(x) \in \mathbb{F}_{16}[x]$ such that $g_1(x)h_1(x) = 1 \mod f(x)$, i.e., $g_1(x)$ is a unit in I. So, $I = \langle 1 + \langle f(x)\rangle\rangle$. If $\gcd(f(x), g_1(x)) = f(x)$, then $g(x) + \langle f(x)\rangle = ug_2(x) + u^2 g_3(x) + u^3 g_4(x) + \langle f(x)\rangle$, which implies $I = \langle u + \langle f(x)\rangle\rangle$. If $\gcd(f(x), g_2(x)) = f(x)$, then $g(x) + \langle f(x)\rangle = u^2 g_3(x) + u^3 g_4(x) + \langle f(x)\rangle$, which shows that $I = \langle u^2 + \langle f(x)\rangle\rangle$. If $\gcd(f(x), g_2(x)) = 1$, then $g(x) \in \langle u + \langle f(x)\rangle\rangle$. Now, $\gcd(f(x), g_3(x)) = f(x)$ implies $g(x) + \langle f(x)\rangle = u^3 g_4(x) + \langle f(x)\rangle$ which shows that $g(x) \in \langle u^3 + \langle f(x)\rangle\rangle$. Again if $\gcd(f(x), g_3(x)) = 1$, then $g(x) \in \langle u^2 + \langle f(x)\rangle\rangle$.

Therefore, combining all cases, we see that $\langle 0\rangle$, $\langle 1\rangle$, $\langle u\rangle$, $\langle u^2\rangle$, and $\langle u^3\rangle$ are only the right $\mathbb{A}$-submodules of $\mathcal{R}_f = \frac{\mathbb{A}[x]}{\langle f\rangle}$. □

The above theorem shows that right $\mathbb{A}$-submodules of $\mathcal{R}_f = \frac{\mathbb{A}[x]}{\langle f\rangle}$ form a chain.

Theorem 3 *Let $x^n - 1 = \prod_{r=1}^{m} f_r$, where f_r's are monic basic irreducible pairwise coprime polynomials in $\mathbb{A}[x]$. Let $\hat{f}_i = \frac{x^n-1}{f_i}$. Then any right $\mathbb{A}$-module in $\frac{\mathbb{A}[x]}{\langle x^n-1\rangle}$ is the sum of the ideals $\langle \hat{f}_i + \langle x^n - 1\rangle\rangle$, $\langle u\hat{f}_i + \langle x^n - 1\rangle\rangle$, $\langle u^2\hat{f}_i + \langle x^n - 1\rangle\rangle$, and $\langle u^3\hat{f}_i + \langle x^n - 1\rangle\rangle$.*

Proof Follows easily by using Theorem 1 and modules of $\frac{\mathbb{A}[x]}{\langle f_i\rangle}$. □

Theorem 4 *Let $\mathcal{C}$ be a cyclic code of length n over $\mathbb{A}$. Then there exists a family of pairwise monic coprime polynomials F_0, F_1, F_2, F_3, F_4 in $\mathbb{F}_{16}[x]$ such that $F_0F_1F_2F_3F_4 = x^n - 1$ and $\mathcal{C} = \langle \hat{F}_1\rangle \oplus \langle u\hat{F}_2\rangle \oplus \langle u^2\hat{F}_3\rangle \oplus \langle u^3\hat{F}_4\rangle$. Moreover, $|\mathcal{C}| = 16^{\sum_{i=0}^{3}(4-i)\ deg(F_{i+1})}$.*

Proof Since $\mathcal{C}$ be a cyclic code of length n over $\mathbb{A}$, it can be written as a direct sum of ideals $\langle u^j\hat{f}_i + \langle x^n - 1\rangle\rangle$ by Theorem 3, where $0 \le j \le 4$ and $1 \le i \le m$. Without loss of generality, let $\mathcal{C} = \langle \hat{f}_{m_1+1}\rangle \oplus \cdots \oplus \langle \hat{f}_{m_1+m_2}\rangle \oplus \langle u\hat{f}_{m_1+m_2+1}\rangle \oplus \cdots \oplus \langle u\hat{f}_{m_1+m_2+m_3}\rangle \oplus \langle u^2\hat{f}_{m_1+m_2+m_3+1}\rangle \oplus \cdots \oplus \langle u^2\hat{f}_{m_1+m_2+m_3+m_4}\rangle \oplus \langle u^3\hat{f}_{m_1+m_2+m_3+m_4+1}\rangle \oplus \cdots \oplus \langle u^3\hat{f}_{m_1+m_2+m_3+m_4+m_5}\rangle$, where, $m_1, m_2, \ldots, m_5 \ge 0$ and $m_1 + m_2 + m_3 + m_4 + 1 \le m$. Consider $m_0 = 0$ and $m_1 + m_2 + m_3 + m_4 + m_5 = m$.

Now we define $F_i = f_{m_0+\cdots+m_i+1} \cdots f_{m_0+\cdots+m_{i+1}}$, $i = 0, 1, \ldots, 4$. These constructed F_i's are pairwise coprime and $x^n - 1 = F_0F_1F_2F_3F_4$. So, $\mathcal{C} = \langle \hat{F}_1\rangle \oplus \langle u\hat{F}_2\rangle \oplus \langle u^2\hat{F}_3\rangle \oplus \langle u^3\hat{F}_4\rangle$.

To find the cardinality of $\mathcal{C}$, we see that $\mathcal{C} = \langle \hat{F}_1\rangle \oplus \langle u\hat{F}_2\rangle \oplus \langle u^2\hat{F}_3\rangle \oplus \langle u^3\hat{F}_4\rangle$. So, $|\mathcal{C}| = |\langle \hat{F}_1\rangle||\langle u\hat{F}_2\rangle||\langle u^2\hat{F}_3\rangle||\langle u^3\hat{F}_4\rangle|$. One can easily verify that $|\langle u^i F_{i+1}\rangle| = 16^{(4-i)\ deg(F_{i+1})}$, for $i = 0, 1, 2, 3$. Which suffices to show that $|\mathcal{C}| = 16^{\sum_{i=0}^{3}(4-i)\ deg(F_{i+1})}$. □

Theorem 5 *Let $\mathcal{C}$ be a cyclic code of length n over $\mathbb{A}$. Then there exist polynomials $h_0(x), h_1(x), h_2(x), h_3(x)$ in $\mathbb{F}_{16}[x]$ such that $\mathcal{C} = \langle h_0(x)\rangle \oplus \langle uh_1(x)\rangle \oplus \langle u^2h_2(x)\rangle \oplus \langle u^3h_3(x)\rangle$ and $h_3(x)|h_2(x)|h_1(x)|h_0(x)|x^n - 1$ in $\mathbb{F}_{16}[x]$.*

Proof By applying the previous theorem, we attain a family of monic pairwise coprime polynomials F_0, F_1, F_2, F_3, F_4 in $\mathbb{F}_{16}[x]$ such that $F_0F_1F_2F_3F_4 = x^n - 1$ and $\mathcal{C} = \langle \hat{F}_1, u\hat{F}_2, u^2\hat{F}_3, u^3\hat{F}_4\rangle$. Now we define a new family of polynomials as

$$h_i(x) = \begin{cases} F_0F_{i+2}\dots F_4, & 0 \le i \le 2, \\ F_0, & i = 3. \end{cases}$$

It is clear that $h_3(x)|h_2(x)|h_1(x)|h_0(x)|x^n - 1$ in $\mathbb{F}_{16}[x]$. Now see that $u^i\hat{F}_{i+1} = u^iF_0\dots F_iF_{i+2}\dots F_4 = u^ih_iF_1\dots F_i$, for $0 \le i \le 3$. So, $\mathcal{C} \subset \langle h_0(x)\rangle \oplus \langle uh_1(x)\rangle \oplus \langle u^2h_2(x)\rangle \oplus \langle u^3h_3(x)\rangle$.

Conversely, we have to show that $\langle h_0(x)\rangle \oplus \langle uh_1(x)\rangle \oplus \langle u^2h_2(x)\rangle \oplus \langle u^3h_3(x)\rangle \subset \mathcal{C}$. We see that $h_0(x) \in \mathcal{C}$. Since, F_1 and F_2 are coprime in $\mathbb{F}_{16}[x]$, there exist $a(x)$ and $b(x)$ in $\mathbb{F}_{16}[x]$ such that $a(x)F_1 + b(x)F_2 = 1$.

So,

$$\begin{aligned} uh_1(x) &= uF_0F_3F_4 \\ &= u(a(x)F_1 + b(x)F_2)F_0F_3F_4 \\ &= ua(x)F_0F_1F_3F_4 + ub(x)F_0F_2F_3F_4 \\ &= ua(x)\hat{F}_2 + ub(x)h_0(x) \in \mathcal{C}. \end{aligned}$$

Similarly, we can show that $u^ih_i(x) \in \mathcal{C}$ for $i = 2$ and 3. Thus, we have $\langle h_0(x)\rangle \oplus \langle uh_1(x)\rangle \oplus \langle u^2h_2(x)\rangle \oplus \langle u^3h_3(x)\rangle \subset \mathcal{C}$.

Hence the theorem is proved. □

Theorem 6 *Let $\mathcal{C}$ be a cyclic code of length n over $\mathbb{A}$. Then $\mathcal{C}$ is generated by F, where $F = \hat{F}_1 + u\hat{F}_2 + u^2\hat{F}_3 + u^3\hat{F}_4$.*

Proof From Theorem 4, we have a family of pairwise monic coprime polynomials F_0, F_1, F_2, F_3, F_4 in $\mathbb{F}_{16}[x]$ such that $F_0F_1F_2F_3F_4 = x^n - 1$ and $\mathcal{C} = \langle\hat{F}_1\rangle \oplus \langle u\hat{F}_2\rangle \oplus \langle u^2\hat{F}_3\rangle \oplus \langle u^3\hat{F}_4\rangle$.

Now, for $i, j \in \{0, 1, 2, 3, 4\}$ and $i \ne j$,

$$\hat{F}_i\hat{F}_j \equiv 0 \mod x^n - 1.$$

It is clear that F_i and $\hat{F}_i$ are coprime for each $i = 1, 2, 3, 4$ in $\mathbb{F}_{16}[x]$. Thus, there exist $g_i(x)$ and $h_i(x)$ in $\mathbb{F}_{16}[x]$ such that for $i \in \{1, 2, 3, 4\}$,

$$g_i(x)F_i + h_i(x)\hat{F}_i = 1 \tag{1}$$

$$\Rightarrow \prod_{i=1}^{k}(g_i(x)F_i + h_i(x)\hat{F}_i) = 1, \quad \text{where } k \in \{1, 2, 3, 4\} \tag{2}$$

$$\Rightarrow a_{k0}F_1F_2\dots F_k + a_{k1}\hat{F}_1F_2\dots F_k + a_{k2}F_1\hat{F}_2\cdots F_k + \cdots + a_{kk}F_1F_2\cdots\hat{F}_k = 1. \tag{3}$$

Putting $k = 3$ in Eq. (3) and multiplying it by $u^3\hat{F}_4$, we get,

$$\begin{aligned} u^3\hat{F}_4a_{30}F_1F_2F_3 &= u^3\hat{F}_4 \\ \Rightarrow u^3a_{30}F_1F_2F_3\hat{F}_4 &= u^3\hat{F}_4. \end{aligned}$$

Now according to our assumption,

$$\begin{aligned} F &= \hat{F}_1 + u\hat{F}_2 + u^2\hat{F}_3 + u^3\hat{F}_4 \\ &\Rightarrow F F_1 F_2 F_3 = u^3 \hat{F}_4 F_1 F_2 F_3 \\ &\Rightarrow F F_1 F_2 F_3 a_{30} = u^3 \hat{F}_4 F_1 F_2 F_3 a_{30} = u^3 a_{30} \hat{F}_4 F_1 F_2 F_3 = u^3 \hat{F}_4 \\ &\Rightarrow u^3 \hat{F}_4 \in \langle F \rangle. \end{aligned}$$

Similarly, we can show that $u^2\hat{F}_3, u\hat{F}_2, \hat{F}_1 \in \langle F \rangle$. Hence $\mathcal{C} = \langle F \rangle$. □

3.1 Gray map

In [4], for the case of the ring $\mathbb{F}_2 + u\mathbb{F}_2$, the isometry map from $a + ub$ to $(b, a + b)$ helps us to extend the Gray map Φ from $\mathbb{A}$ to $\mathbb{F}_{16}^4$ such that,

$$\Phi(a + ub + u^2c + u^3d) \mapsto (d, c + d, b + c + d, a + b + c + d)$$

preserves Lee weight and Hamming weight.

Thus for any element $x = a + ub + u^2b + u^3b$ in $\mathbb{A}$, extended Lee weight is defined as

$$w_L(x) = w_H(d) + w_H(c + d) + w_H(b + c + d) + w_H(a + b + c + d),$$

where $w_L(\cdot)$ and $w_H(\cdot)$ stand for Lee weight and Hamming weight respectively. If $\bar{x} = (x_0, x_1, x_2, \ldots x_{n-1}) \in \mathbb{A}^n$, then Lee weight of $\bar{x}$ is defined as,

$$w_L(\bar{x}) = w_L(x_0) + w_L(x_1) + w_L(x_2) + \cdots + w_L(x_{n-1}).$$

Now if $\mathcal{C}$ is a linear code over $\mathbb{A}$, then the Lee distance of the code is defined by

$$d_L = \min\{w_L(\bar{x}) | \bar{x} \in \mathcal{C}\}.$$

Theorem 7 *Let n be odd and $\mathcal{C}$ be a linear code over $\mathbb{A}$ of length n with size M and minimum Lee distance d. Then $\Phi(\mathcal{C})$ is a code of length $4n$ with size M and minimum Hamming distance d.*

Now, we show some cyclic codes of length 3 and 7 and their Gray maps described in the following examples. We have found that the image of a cyclic code over $\mathbb{A}$ is a linear code over $\mathbb{F}_{16}$.

Example 1 Consider the factorization $x^3 - 1 = (x+1)(x+w^2+w)(x+w^2+w+1)$ over $\mathbb{F}_{16}$. Let $f_1 = x + 1$, $f_2 = x + w^2 + w$ and $f_3 = x + w^2 + w + 1$. Images of some cyclic codes of length 3 over $\mathbb{A}$ under the Gray map are given below.

Generators of cyclic codes	Images under Φ
$\langle f_1, uf_2\rangle$	[12, 11, 2]
$\langle u^3 f_2, f_2 f_3\rangle$	[12, 6, 3]
$\langle f_1, uf_1, u^2 f_2 f_3\rangle$	[12, 10, 2]
$\langle uf_2, u^2 f_2, u^3 f_3\rangle$	[12, 8, 2]

Example 2 Consider the factorization $x^7 - 1 = (x + 1)(x^3 + x + 1)(x^3 + x^2 + 1)$ over $\mathbb{F}_{16}$. Let $f_1 = x + 1$, $f_2 = x^3 + x + 1$ and $f_3 = x^3 + x^2 + 1$. Images of some cyclic codes of length 7 over $\mathbb{A}$ under the Gray map are given below.

Generators of cyclic codes	Images under Φ
$\langle uf_2, u^2 f_2 f_3\rangle$	[28, 14, 3]
$\langle u^2 f_1, uf_2, u^3 f_2 f_3\rangle$	[28, 25, 2]
$\langle u^3 f_2, u^2 f_3, f_2 f_3\rangle$	[28, 16, 3]

4 Conclusion

In this paper, we have furnished cyclic codes over a non-commutative matrix ring and exhibited that the right-$\mathbb{A}$ modules form a chain. Thus, the structure of the cyclic codes over $\mathbb{A}$ follows from the structure of the cyclic codes over the finite chain rings. Moreover, we have defined a Gray map which outlines a correspondence between $\mathcal{M}_4(\mathbb{F}_2)$ to the field of order 16. However, a purposeful research venture can be carried forward on the self-dual codes over $\mathcal{M}_4(\mathbb{F}_2)$. Further, the structure of the cyclic codes over $\mathcal{M}_3(\mathbb{F}_2)$ due to [9] are yet to be dealt with and can be thought of a challenging open problem.

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